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Chapter 5 - Exact First-Order Equations

This chapter develops a systematic method for differential equations that hide the total derivative of a two-variable function.

The method has two central tasks:

1. decide whether the equation is exact
2. reconstruct the function whose differential appears in the equation

When an equation is not exact, a carefully chosen integrating factor may make it exact.

We will work step by step. In every reconstruction, we will state which variable is held constant, preserve the unknown one-variable function created by partial integration, and compare coefficients one line at a time.

The main questions are:

- what does exactness mean?
- why do cross-partial derivatives provide a test?
- how is the hidden potential function recovered?
- when should we integrate the dx coefficient or the dy coefficient?
- how does an initial condition select one level curve?
- how can a factor depending only on x or only on y restore exactness?

The central idea is:

an exact differential equation describes motion along a level curve of one underlying potential function.

Exactness As A Hidden Total Derivative

The Goal Of This Block

The goal is to connect differential form with the multivariable chain rule.

This explains both the meaning of an exact equation and the form of its solution.

Start With A Two-Variable Function

Let:

$$\Phi = \Phi(x, y)$$

be a differentiable function of two variables.

Its total differential is:

$$d\Phi = \Phi_x dx + \Phi_y dy$$

The notation means:

$$\Phi_x = \frac{\partial \Phi}{\partial x} \quad \text{and} \quad \Phi_y = \frac{\partial \Phi}{\partial y}$$

If a differential equation has the form:

$$M(x, y) dx + N(x, y) dy = 0$$

and there is a function Φ satisfying:

$$\Phi_x = M \quad \text{and} \quad \Phi_y = N,$$

then the equation is **exact**.

In that case:

$$M dx + N dy = d\Phi$$

and the differential equation becomes:

$$d\Phi = 0$$

Why The Solution Is A Level Curve

Along a solution $y = y(x)$, the chain rule gives:

$$\frac{d}{dx}\Phi(x, y(x)) = \Phi_x + \Phi_y y'$$

Replace the partial derivatives by M and N :

$$\frac{d}{dx}\Phi(x, y(x)) = M + Ny'$$

The differential equation:

$$M dx + N dy = 0$$

becomes, after division by dx :

$$M + Ny' = 0$$

The expression $M + Ny'$ is the derivative of the potential along the solution, and the differential equation says that this same expression equals zero. Substitute that value into the derivative relation:

$$\frac{d}{dx}\Phi(x, y(x)) = 0$$

Integrating with respect to x gives:

$$\boxed{\Phi(x, y) = C}$$

The practical meaning is:

a solution moves through the (x, y) -plane without changing the value of the potential Φ .

A Potential Built First

Consider the function:

$$\Phi(x, y) = xy^2 + \sin x$$

Differentiate with respect to x , holding y constant:

$$\begin{aligned}\Phi_x &= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial x}(\sin x) \\ &= y^2 + \cos x.\end{aligned}$$

Differentiate with respect to y , holding x constant:

$$\begin{aligned}\Phi_y &= \frac{\partial}{\partial y}(xy^2) + \frac{\partial}{\partial y}(\sin x) \\ &= 2xy + 0 \\ &= 2xy.\end{aligned}$$

Recall the total-differential formula:

$$d\Phi = \Phi_x dx + \Phi_y dy$$

Substitute the two partial derivatives just calculated:

$$\begin{aligned}d\Phi &= \Phi_x dx + \Phi_y dy \\ &= (y^2 + \cos x) dx + 2xy dy.\end{aligned}$$

The equation:

$$(y^2 + \cos x) dx + 2xy dy = 0$$

has exactly the same left side as $d\Phi$. Therefore the equation can be rewritten as:

$$d\Phi = 0$$

A function with zero differential is constant along a solution, so integrate to obtain:

$$\Phi(x, y) = C$$

Finally, substitute the original function $\Phi(x, y) = xy^2 + \sin x$:

$xy^2 + \sin x = C$

Verification From The Implicit Solution

Start from:

$$xy^2 + \sin x = C$$

Differentiate every term with respect to x .

For xy^2 , use the product rule and remember that $y = y(x)$:

$$\frac{d}{dx}(xy^2) = y^2 + x(2yy')$$

Also:

$$\frac{d}{dx}(\sin x) = \cos x$$

and:

$$\frac{dC}{dx} = 0$$

Place the three differentiated terms into the equation:

$$y^2 + 2xyy' + \cos x = 0$$

Reorder the terms so that the part containing y' is visible:

$$(y^2 + \cos x) + 2xyy' = 0$$

Replace y' by dy/dx :

$$(y^2 + \cos x) + 2xy \frac{dy}{dx} = 0$$

Multiply every term by dx :

$$(y^2 + \cos x) dx + 2xy dy = 0$$

This is exactly the original differential equation.

Block 1 Summary

An equation:

$$M dx + N dy = 0$$

is exact when:

$$M = \Phi_x \quad \text{and} \quad N = \Phi_y$$

for one potential function Φ . Its solutions lie on level curves:

$$\Phi(x, y) = C$$

Testing An Equation For Exactness

The Goal Of This Block

The goal is to decide whether a potential function exists before attempting to reconstruct it.

Deriving The Cross-Partial Test

Suppose:

$$M = \Phi_x \quad \text{and} \quad N = \Phi_y$$

Differentiate M with respect to y :

$$M_y = (\Phi_x)_y = \Phi_{xy}$$

Differentiate N with respect to x :

$$N_x = (\Phi_y)_x = \Phi_{yx}$$

When the relevant partial derivatives are continuous:

$$\Phi_{xy} = \Phi_{yx}$$

Therefore an exact equation must satisfy:

$$\boxed{M_y = N_x}$$

On a suitable simply connected domain, this equality is also sufficient. For the introductory examples in this chapter, the working domains will be rectangles, half-planes, or the whole plane unless stated otherwise.

The Four-Step Test

For:

$$M(x, y) dx + N(x, y) dy = 0,$$

use these steps.

Step 1: Identify the coefficients. Write M and N separately.

Step 2: Differentiate across the variables. Calculate:

$$M_y \quad \text{and} \quad N_x$$

Step 3: Compare the complete expressions. Do not compare only one matching term.

Step 4: State the domain conclusion. If the derivatives agree on a suitable domain, the equation is exact there. If they do not agree, the displayed form is not exact.

Worked Exactness Test

Test:

$$(2xy + e^x \cos y) dx + (x^2 - e^x \sin y + 4y^3) dy = 0$$

Step 1: Identify M and N .

$$M(x, y) = 2xy + e^x \cos y$$

$$N(x, y) = x^2 - e^x \sin y + 4y^3$$

Step 2: Compute M_y . Hold x constant:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(2xy) + \frac{\partial}{\partial y}(e^x \cos y) \\ &= 2x - e^x \sin y. \end{aligned}$$

Step 3: Compute N_x . Hold y constant:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial x}(e^x \sin y) + \frac{\partial}{\partial x}(4y^3) \\ &= 2x - e^x \sin y + 0 \\ &= 2x - e^x \sin y. \end{aligned}$$

Step 4: Compare.

$$M_y = N_x$$

Both coefficient functions and their first partial derivatives are continuous on the plane. Therefore the equation is exact on $\mathbb{R}^2$.

A Nonexact Near Miss

Test:

$$(2xy + e^y) dx + (x^2 + x + 3y^2) dy = 0$$

Identify:

$$M(x, y) = 2xy + e^y$$

and:

$$N(x, y) = x^2 + x + 3y^2$$

Differentiate M with respect to y :

$$M_y = 2x + e^y.$$

Differentiate N with respect to x :

$$N_x = 2x + 1.$$

The expressions are not equal in general:

$$2x + e^y \neq 2x + 1$$

Therefore the displayed equation is not exact.

The fact that the terms $2x$ match is not enough. The full cross-partial expressions must agree.

Exactness Belongs To A Form And A Domain

Multiplying a differential form by a nonzero function preserves its local solution curves, but it may change exactness.

Similarly, an expression containing $1/x$, $1/y$, or a logarithm may be exact only on regions that avoid its singularities.

For this reason, say:

the displayed differential form is exact on the stated domain.

This is more precise than saying that an equation is exact without reference to its representation or domain.

Block 2 Summary

For:

$$M dx + N dy = 0,$$

calculate:

$$M_y \quad \text{and} \quad N_x$$

Equality on a suitable domain signals exactness. Inequality means that the displayed form is not exact.

Reconstructing The Potential From M

The Goal Of This Block

The goal is to recover $\Phi(x, y)$ from an exact equation without skipping the one-variable correction term.

The Reconstruction Method

Suppose the equation is exact:

$$M dx + N dy = 0$$

and:

$$\Phi_x = M, \quad \Phi_y = N$$

To reconstruct Φ from M :

1. integrate M with respect to x , holding y constant
2. add an unknown function $h(y)$
3. differentiate the result with respect to y
4. set that derivative equal to N
5. solve for $h'(y)$ and then $h(y)$
6. write $\Phi(x, y) = C$

The term $h(y)$ is essential. When integrating with respect to x , every function of y behaves like a constant.

Complete Worked Example

Solve:

$$(2xy + e^x \cos y) dx + (x^2 - e^x \sin y + 4y^3) dy = 0$$

Before reconstructing the potential, verify exactness locally. Here:

$$M = 2xy + e^x \cos y$$

and:

$$N = x^2 - e^x \sin y + 4y^3$$

Their cross-partials are:

$$\begin{aligned}M_y &= \frac{\partial}{\partial y}(2xy + e^x \cos y) \\&= 2x - e^x \sin y\end{aligned}$$

and:

$$\begin{aligned}N_x &= \frac{\partial}{\partial x}(x^2 - e^x \sin y + 4y^3) \\&= 2x - e^x \sin y\end{aligned}$$

Since $M_y = N_x$ on the plane, the equation is exact and a potential Φ can be reconstructed.

Step 1: Use $\Phi_x = M$.

$$\Phi_x = 2xy + e^x \cos y$$

Step 2: Integrate with respect to x . During this integration, y is held constant.

Integrate the first term:

$$\int 2xy \, dx = y \int 2x \, dx = x^2 y$$

Integrate the second term:

$$\int e^x \cos y \, dx = \cos y \int e^x \, dx = e^x \cos y$$

Add the unknown function of y :

$$\Phi(x, y) = x^2 y + e^x \cos y + h(y)$$

Step 3: Differentiate this result with respect to y .

$$\begin{aligned}\Phi_y &= \frac{\partial}{\partial y}(x^2 y) + \frac{\partial}{\partial y}(e^x \cos y) + h'(y) \\&= x^2 - e^x \sin y + h'(y).\end{aligned}$$

Step 4: Compare with the original N . The original coefficient of dy is:

$$N = x^2 - e^x \sin y + 4y^3$$

Set $\Phi_y = N$:

$$x^2 - e^x \sin y + h'(y) = x^2 - e^x \sin y + 4y^3$$

Subtract the common terms from both sides:

$$h'(y) = 4y^3$$

Step 5: Integrate for $h(y)$.

$$\begin{aligned} h(y) &= \int 4y^3 dy \\ &= y^4. \end{aligned}$$

An additive constant inside h is unnecessary because the final equation already contains an arbitrary constant.

Step 6: Assemble the potential.

$$\boxed{\Phi(x, y) = x^2 y + e^x \cos y + y^4}$$

Step 7: Write the implicit solution. The reconstructed derivatives satisfy $\Phi_x = M$ and $\Phi_y = N$. Hence the left side of the original equation is $d\Phi$, so the equation becomes:

$$d\Phi = 0$$

Integrating gives:

$$\Phi(x, y) = C$$

Substitute the potential from Step 6:

$$x^2y + e^x \cos y + y^4 = C$$

Final Potential Check

Differentiate the reconstructed potential both ways.

With respect to x :

$$\begin{aligned}\Phi_x &= 2xy + e^x \cos y + 0 \\ &= M.\end{aligned}$$

With respect to y :

$$\begin{aligned}\Phi_y &= x^2 - e^x \sin y + 4y^3 \\ &= N.\end{aligned}$$

Both coefficients have been recovered, so the potential is correct.

The Most Common Reconstruction Error

The incorrect step is:

$$\Phi = \int M(x, y) dx + C$$

when C is treated as a numerical constant.

The correct step is:

$$\Phi = \int M(x, y) dx + h(y)$$

because a function depending only on y has zero partial derivative with respect to x .

The comparison with N is what determines that missing function.

Block 3 Summary

Integrate M with respect to x , append $h(y)$, differentiate with respect to y , and compare with N .

Do not omit the one-variable correction function. It carries all potential terms that are invisible to Φ_x .

Reconstructing From N Instead

The Goal Of This Block

The goal is to use the direction of integration that produces the cleanest calculation.

The Alternative Route

Starting from:

$$\Phi_y = N,$$

we may integrate with respect to y :

$$\Phi(x, y) = \int N(x, y) dy + k(x)$$

Now x is held constant, so the missing term may be any function $k(x)$.

Then:

1. differentiate the result with respect to x
2. compare with M
3. determine $k'(x)$
4. integrate to find $k(x)$

Worked Example Using N

Solve:

$$(2x \sin y + 3x^2) dx + (x^2 \cos y + 2y) dy = 0$$

Step 1: Test exactness. Identify:

$$M = 2x \sin y + 3x^2$$

$$N = x^2 \cos y + 2y$$

Differentiate:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y} (2x \sin y + 3x^2) \\ &= 2x \cos y \end{aligned}$$

and:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x} (x^2 \cos y + 2y) \\ &= 2x \cos y \end{aligned}$$

The cross-partials agree, so the equation is exact on the plane.

Step 2: Integrate N with respect to y . Hold x constant:

$$\Phi(x, y) = \int (x^2 \cos y + 2y) dy + k(x)$$

Integrate one term at a time:

$$\int x^2 \cos y dy = x^2 \sin y$$

$$\int 2y dy = y^2$$

Therefore:

$$\Phi(x, y) = x^2 \sin y + y^2 + k(x)$$

Step 3: Differentiate with respect to x .

$$\begin{aligned}\Phi_x &= 2x \sin y + 0 + k'(x) \\ &= 2x \sin y + k'(x).\end{aligned}$$

Step 4: Compare with M . Set $\Phi_x = M$:

$$2x \sin y + k'(x) = 2x \sin y + 3x^2$$

Subtract $2x \sin y$ from both sides:

$$k'(x) = 3x^2$$

Step 5: Integrate for $k(x)$.

$$\begin{aligned}k(x) &= \int 3x^2 dx \\ &= x^3.\end{aligned}$$

Step 6: Write the solution. The potential is:

$$\Phi = x^2 \sin y + y^2 + x^3$$

Because its partial derivatives reproduce M and N , the differential form is $d\Phi$. The equation is therefore:

$$d\Phi = 0$$

Integrating gives $\Phi = C$. Substituting the displayed potential produces the implicit solution:

$$\boxed{x^2 \sin y + y^2 + x^3 = C}$$

How To Choose The Easier Direction

Integrate M with respect to x when its x -antiderivative is simpler.

Integrate N with respect to y when its y -antiderivative is simpler.

For example:

- $x^2 \cos y$ is easy to integrate with respect to y
- $e^x \cos y$ is easy to integrate with respect to x
- a term may be awkward in one direction but immediate in the other

Both routes must produce potentials differing by at most an additive constant.

Block 4 Summary

The potential may be reconstructed from either coefficient:

$$\Phi = \int M dx + h(y)$$

or:

$$\Phi = \int N dy + k(x)$$

Choose the cleaner antiderivative, then compare the unused partial derivative with the untouched coefficient.

Slope Form And Initial Conditions

The Goal Of This Block

The goal is to move from slope form to differential form and use an initial condition to select one level curve.

Converting Slope Form

Suppose:

$$y' = -\frac{M(x, y)}{N(x, y)}$$

where $N \neq 0$.

Multiply by N :

$$Ny' = -M$$

Move all terms to one side:

$$M + Ny' = 0$$

Replace y' by dy/dx and multiply by dx :

$$M dx + N dy = 0$$

Only after this conversion should the cross-partial test be applied.

Worked Slope-Form Example

Solve:

$$y' = -\frac{2x + y}{x + e^y}$$

on a region where $x + e^y \neq 0$.

Step 1: Convert to differential form. Multiply by $x + e^y$:

$$(x + e^y)y' = -(2x + y)$$

Move the right side to the left:

$$(2x + y) + (x + e^y)y' = 0$$

Replace y' by dy/dx :

$$(2x + y) + (x + e^y)\frac{dy}{dx} = 0$$

Multiply every term by dx :

$$(2x + y) dx + (x + e^y) dy = 0$$

The coefficient multiplying dx is M , and the coefficient multiplying dy is N . Therefore:

$$M = 2x + y, \quad N = x + e^y$$

Step 2: Test exactness.

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(2x + y) \\ &= 1 \end{aligned}$$

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(x + e^y) \\ &= 1 \end{aligned}$$

The equation is exact.

Step 3: Integrate M with respect to x . Hold y constant:

$$\begin{aligned} \Phi &= \int (2x + y) dx + h(y) \\ &= x^2 + xy + h(y). \end{aligned}$$

Step 4: Differentiate with respect to y .

$$\Phi_y = x + h'(y)$$

Step 5: Compare with N .

$$x + h'(y) = x + e^y$$

Subtract x :

$$h'(y) = e^y$$

Integrate:

$$h(y) = e^y$$

Step 6: Write the solution. Substitute $h(y) = e^y$ into the expression from Step 3:

$$\Phi(x, y) = x^2 + xy + e^y$$

The exact equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so:

$$x^2 + xy + e^y = C$$

Worked Initial-Value Problem

Solve:

$$(2x \sin y + 3x^2) dx + (x^2 \cos y + 2y) dy = 0, \quad y(0) = 1$$

First establish the potential within this example. Take:

$$\Phi(x, y) = x^2 \sin y + x^3 + y^2$$

Differentiate it with respect to x :

$$\Phi_x = 2x \sin y + 3x^2$$

This matches the coefficient of dx . Differentiate it with respect to y :

$$\Phi_y = x^2 \cos y + 2y$$

This matches the coefficient of dy . Therefore the differential equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so the solution family is:

$$x^2 \sin y + x^3 + y^2 = C$$

Apply the initial point $(0, 1)$:

$$(0)^2 \sin(1) + (0)^3 + (1)^2 = C$$

Simplify each term:

$$0 + 0 + 1 = C$$

Hence:

$$C = 1$$

The initial-value solution is:

$$\boxed{x^2 \sin y + x^3 + y^2 = 1}$$

At the initial point, the coefficient of dy is:

$$N(0, 1) = 0^2 \cos(1) + 2(1) = 2$$

Since this is nonzero, the implicit relation defines a differentiable local branch $y(x)$ through $(0, 1)$.

Why Implicit Form Is Often Best

The equation:

$$x^2 \sin y + x^3 + y^2 = 1$$

does not have a convenient elementary rearrangement for y .

That is not an incomplete answer. The implicit relation:

- identifies the correct level curve
- satisfies the initial condition
- can be differentiated to recover the differential equation
- can support numerical evaluation of y when needed

Block 5 Summary

Convert slope form to $M dx + N dy = 0$, test exactness, reconstruct the potential, and then use the initial condition to determine C .

The solution naturally remains implicit when algebraic isolation of y is unhelpful.

Why Integrating Factors Work

The Goal Of This Block

The goal is to understand an integrating factor as a multiplier that repairs the failed cross-partial condition.

The Basic Idea

Suppose:

$$M dx + N dy = 0$$

is not exact.

We seek a nonzero function μ such that:

$$\mu M dx + \mu N dy = 0$$

is exact.

The function μ is called an **integrating factor**.

The repaired cross-partial condition is:

$$(\mu M)_y = (\mu N)_x$$

Finding a completely general $\mu(x, y)$ is usually difficult. Two useful special cases occur when μ depends only on x or only on y .

Deriving A Factor Depending Only On x

Assume:

$$\mu = \mu(x)$$

Because μ does not depend on y :

$$(\mu M)_y = \mu M_y$$

Use the product rule on the other side:

$$(\mu N)_x = \mu' N + \mu N_x$$

Exactness requires:

$$\mu M_y = \mu' N + \mu N_x$$

Move the μN_x term to the left:

$$\mu(M_y - N_x) = \mu' N$$

Divide by μN where both are nonzero:

$$\frac{\mu'}{\mu} = \frac{M_y - N_x}{N}$$

If the right side simplifies to a function $R(x)$ of x alone, then:

$$\frac{\mu'}{\mu} = R(x)$$

Integrate both sides with respect to x :

$$\int \frac{\mu'}{\mu} dx = \int R(x) dx$$

Since the derivative of $\ln |\mu|$ is μ' / μ :

$$\ln |\mu| = \int R(x) dx + K$$

Exponentiate both sides:

$$|\mu| = e^K \exp \left(\int R(x) dx \right)$$

The factor e^K is a positive constant, and either sign of μ is allowed. Since multiplying an integrating factor by any nonzero constant produces another integrating factor, choose that constant to be 1. A convenient integrating factor is therefore:

$$\mu(x) = \exp \left(\int R(x) dx \right)$$

A nonzero constant multiple of this choice is equally valid.

Deriving A Factor Depending Only On y

Now assume:

$$\mu = \mu(y)$$

Use the product rule with respect to y :

$$(\mu M)_y = \mu' M + \mu M_y$$

Since μ does not depend on x :

$$(\mu N)_x = \mu N_x$$

Set the expressions equal:

$$\mu' M + \mu M_y = \mu N_x$$

Move the μM_y term to the right:

$$\mu' M = \mu(N_x - M_y)$$

Divide by μM where permitted:

$$\frac{\mu'}{\mu} = \frac{N_x - M_y}{M}$$

If the right side is a function $S(y)$ of y alone, then:

$$\frac{\mu'}{\mu} = S(y)$$

Integrate both sides with respect to y :

$$\int \frac{\mu'}{\mu} dy = \int S(y) dy$$

Since the derivative of $\ln |\mu|$ is μ' / μ :

$$\ln |\mu| = \int S(y) dy + K$$

Exponentiate:

$$|\mu| = e^K \exp \left(\int S(y) dy \right)$$

As in the x -only case, omit the arbitrary nonzero constant multiple and choose:

$$\mu(y) = \exp \left(\int S(y) dy \right)$$

Notice the order of the numerator:

$$N_x - M_y$$

for a y -only factor, but:

$$M_y - N_x$$

for an x -only factor.

A Decision Rule

For a nonexact form, calculate:

$$R(x, y) = \frac{M_y - N_x}{N}$$

and:

$$S(x, y) = \frac{N_x - M_y}{M}$$

Then:

- if R depends only on x , try $\mu(x) = e^{\int R dx}$
- if S depends only on y , try $\mu(y) = e^{\int S dy}$
- if neither simplification occurs, these one-variable tests do not provide an integrating factor

Always verify exactness after multiplying. The formula is a route to a candidate, not a replacement for the final check.

Block 6 Summary

An integrating factor changes a nonexact form into an exact one.

Useful tests are:

$$\frac{M_y - N_x}{N} = R(x) \implies \mu(x) = e^{\int R(x) dx}$$

and:

$$\frac{N_x - M_y}{M} = S(y) \implies \mu(y) = e^{\int S(y) dy}$$

A Complete x-Only Integrating-Factor Example

The Goal Of This Block

The goal is to carry the entire process from a failed exactness test to a verified implicit solution.

The Problem

Solve:

$$2x(y + 1) dx + (1 + 2ye^{-x^2}) dy = 0$$

Step 1: Test The Original Form

Identify:

$$M = 2x(y + 1)$$

$$N = 1 + 2ye^{-x^2}$$

Differentiate M with respect to y :

$$M_y = 2x$$

Differentiate N with respect to x :

$$\begin{aligned} N_x &= 2y \frac{d}{dx}(e^{-x^2}) \\ &= 2y(-2x)e^{-x^2} \\ &= -4xye^{-x^2}. \end{aligned}$$

Since:

$$2x \neq -4xye^{-x^2}$$

in general, the original form is not exact.

Step 2: Test For An x -Only Factor

Compute:

$$\frac{M_y - N_x}{N}$$

Substitute the derivatives:

$$\frac{2x - (-4xye^{-x^2})}{1 + 2ye^{-x^2}}$$

Remove the double negative:

$$\frac{2x + 4xye^{-x^2}}{1 + 2ye^{-x^2}}$$

Factor $2x$ from the numerator:

$$\frac{2x(1 + 2ye^{-x^2})}{1 + 2ye^{-x^2}}$$

Cancel the common factor where it is nonzero:

$$\boxed{\frac{M_y - N_x}{N} = 2x}$$

The result depends only on x .

Step 3: Construct The Integrating Factor

Use:

$$\mu(x) = \exp\left(\int 2x \, dx\right)$$

Evaluate the integral:

$$\int 2x \, dx = x^2$$

Therefore:

$$\boxed{\mu(x) = e^{x^2}}$$

Step 4: Multiply Every Term

Multiply the complete differential equation by e^{x^2} :

$$e^{x^2} [2x(y + 1)] \, dx + e^{x^2} (1 + 2ye^{-x^2}) \, dy = 0$$

Simplify the dy coefficient:

$$\begin{aligned} e^{x^2} (1 + 2ye^{-x^2}) &= e^{x^2} + 2ye^{x^2} e^{-x^2} \\ &= e^{x^2} + 2y. \end{aligned}$$

The transformed equation is:

$$\boxed{2xe^{x^2}(y + 1) \, dx + (e^{x^2} + 2y) \, dy = 0}$$

Step 5: Verify Exactness

Let:

$$\widetilde{M} = 2xe^{x^2}(y + 1)$$

$$\widetilde{N} = e^{x^2} + 2y$$

Then:

$$\widetilde{M}_y = 2xe^{x^2}$$

and:

$$\widetilde{N}_x = 2xe^{x^2}$$

The transformed form is exact.

The quotient in Step 2 was simplified only where $1 + 2ye^{-x^2} \neq 0$. However, the candidate e^{x^2} is defined and nonzero everywhere, and the direct cross-partial check above works on the whole plane. That verification extends the transformed exact equation across the points where the quotient itself was undefined.

Step 6: Reconstruct The Potential

Integrate $\widetilde{M}$ with respect to x , holding y constant:

$$\Phi = \int 2xe^{x^2}(y + 1) dx + h(y)$$

Factor out $y + 1$:

$$\Phi = (y + 1) \int 2xe^{x^2} dx + h(y)$$

Since:

$$\frac{d}{dx}(e^{x^2}) = 2xe^{x^2},$$

we obtain:

$$\Phi = e^{x^2}(y + 1) + h(y)$$

Differentiate with respect to y :

$$\Phi_y = e^{x^2} + h'(y)$$

Compare with $\tilde{N}$:

$$e^{x^2} + h'(y) = e^{x^2} + 2y$$

Subtract e^{x^2} :

$$h'(y) = 2y$$

Integrate:

$$h(y) = y^2$$

Substitute $h(y) = y^2$ into $\Phi = e^{x^2}(y + 1) + h(y)$:

$$\Phi(x, y) = e^{x^2}(y + 1) + y^2$$

Step 7: State And Check The Solution

The transformed exact equation has coefficients Φ_x and Φ_y , so its left side is $d\Phi$. Therefore:

$$d\Phi = 0$$

Integrate to obtain $\Phi = C$, then substitute the potential from Step 6. The implicit solution is:

$$e^{x^2}(y+1) + y^2 = C$$

Differentiate the potential:

$$\Phi_x = 2xe^{x^2}(y+1)$$

$$\Phi_y = e^{x^2} + 2y$$

These are exactly the transformed coefficients. Since e^{x^2} is never zero, multiplying by it did not introduce or remove points through a zero of the integrating factor.

Block 7 Summary

The full integrating-factor workflow is:

1. fail the original exactness test
2. simplify the one-variable quotient
3. integrate to obtain μ
4. multiply every term
5. verify exactness again
6. reconstruct the potential
7. check the result and domain

A Complete y-Only Integrating-Factor Example

The Goal Of This Block

The goal is to repeat the method when the repairing factor depends only on y .

The Problem And Original Test

Solve:

$$(2x + 1) dx + (x^2 + x + e^{-y}) dy = 0$$

Identify:

$$M = 2x + 1$$

$$N = x^2 + x + e^{-y}$$

Compute:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(2x + 1) \\ &= 0 \end{aligned}$$

and:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(x^2 + x + e^{-y}) \\ &= 2x + 1 \end{aligned}$$

The form is not exact.

Step 1: Test For A y -Only Factor

Compute:

$$\frac{N_x - M_y}{M}$$

Substitute:

$$\frac{(2x + 1) - 0}{2x + 1}$$

Simplify where $2x + 1 \neq 0$:

$$\frac{N_x - M_y}{M} = 1$$

The result is a function of y alone because a constant is allowed.

Step 2: Construct And Apply The Factor

Use:

$$\mu(y) = \exp\left(\int 1 \, dy\right)$$

Therefore:

$$\mu(y) = e^y$$

Multiply every term by e^y :

$$e^y(2x + 1) \, dx + e^y(x^2 + x + e^{-y}) \, dy = 0$$

Simplify:

$$e^y e^{-y} = 1$$

so the transformed equation is:

$$e^y(2x + 1) \, dx + (e^y(x^2 + x) + 1) \, dy = 0$$

Step 3: Verify And Solve

Set:

$$\tilde{M} = e^y(2x + 1)$$

$$\tilde{N} = e^y(x^2 + x) + 1$$

Differentiate:

$$\widetilde{M}_y = e^y(2x + 1)$$

$$\widetilde{N}_x = e^y(2x + 1)$$

The transformed equation is exact.

The quotient used to discover the factor required $2x + 1 \neq 0$. The final factor e^y , however, is defined and nonzero for every (x, y) , and the cross-partials above agree even when $x = -\frac{1}{2}$. Thus the verified transformed equation is exact on the whole plane.

Integrate $\widetilde{M}$ with respect to x , holding y constant:

$$\begin{aligned}\Phi &= \int e^y(2x + 1) dx + h(y) \\ &= e^y(x^2 + x) + h(y).\end{aligned}$$

Differentiate with respect to y :

$$\Phi_y = e^y(x^2 + x) + h'(y)$$

Compare with $\widetilde{N}$:

$$e^y(x^2 + x) + h'(y) = e^y(x^2 + x) + 1$$

Cancel the common term:

$$h'(y) = 1$$

Integrate:

$$h(y) = y$$

Substitute $h(y) = y$ into the reconstructed expression for Φ :

$$\Phi(x, y) = e^y(x^2 + x) + y$$

The transformed exact equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so:

$$e^y(x^2 + x) + y = C$$

Integrating-Factor Cautions

An integrating factor may be valid only on part of the plane.

For example, factors such as:

$$\frac{1}{x}, \quad \frac{1}{y^2}, \quad |x|^a$$

require explicit domain restrictions.

Multiplication by a finite nonzero factor preserves the local solution curves. A factor that is zero, undefined, or obtained after division by a vanishing coefficient requires additional checking in the original equation.

Integrating factors are also not unique. If μ works, then any nonzero constant multiple $K\mu$ works as well.

Block 8 Summary

For a y -only factor, simplify:

$$\frac{N_x - M_y}{M}$$

to a function of y , integrate it, multiply the whole equation, and verify exactness before reconstructing the potential.

A Step-By-Step Checklist

For An Equation That May Already Be Exact

1. Put the equation in the form $M dx + N dy = 0$.
2. Write M and N on separate lines.
3. Calculate M_y and N_x .
4. Compare the complete expressions and state the domain.
5. Integrate the easier coefficient in the appropriate variable.
6. Add $h(y)$ or $k(x)$.
7. Differentiate in the other variable.
8. Compare with the untouched coefficient.
9. Solve for the missing one-variable function.
10. Write $\Phi = C$ and verify both partial derivatives.

For A Nonexact Equation

1. Confirm that $M_y \neq N_x$.
2. Calculate $(M_y - N_x)/N$.
3. Check whether it depends only on x .
4. Calculate $(N_x - M_y)/M$.
5. Check whether it depends only on y .
6. Construct the available integrating factor.
7. Record its domain.
8. Multiply both coefficients, not just one term.
9. Retest exactness using the transformed coefficients.
10. Solve the resulting exact equation.
11. Verify the result against the original domain and equation.

Common Mistakes

Comparing the wrong derivatives. The test is:

$$M_y = N_x$$

not $M_x = N_y$.

Treating a partial-integration constant as a number. Use $h(y)$ after integrating with respect to x , and $k(x)$ after integrating with respect to y .

Integrating both coefficients and adding the results. This usually double-counts shared terms. Integrate one coefficient, then use the other to determine the missing function.

Applying an integrating-factor formula with both variables remaining. The x -only formula requires an expression depending only on x . The y -only formula requires an expression depending only on y .

Failing to retest after multiplication. A sign error in the quotient or integral is exposed immediately by the final cross-partial check.

Block 9 Summary

Exact-equation work is reliable when every stage has a check:

- cross-partials check the classification
- coefficient comparison checks the missing function
- differentiating Φ checks the reconstruction
- retesting checks the integrating factor
- the original equation and domain check the final result

Original Practice Set

Practice Problems

Problem 1: Test Exactness. Determine whether:

$$(y^2 + x) dx + (2xy + e^y) dy = 0$$

is exact.

Problem 2: Reconstruct From M . Solve:

$$(2xy + \cos x) dx + (x^2 + 3y^2) dy = 0$$

Problem 3: Reconstruct From N . Solve:

$$(e^x + y) dx + (x + 2y) dy = 0$$

by beginning with the coefficient of dy .

Problem 4: Convert From Slope Form. Convert and solve:

$$y' = -\frac{y + 2x}{x + 2y}$$

on a region where the denominator is nonzero.

Problem 5: Apply An Initial Condition. Solve:

$$(e^x + y) dx + (x + 2y) dy = 0, \quad y(0) = 1$$

Problem 6: Find An x -Only Integrating Factor. For $x \neq -1$, find an integrating factor depending only on x for:

$$\left(2 + \frac{3y}{1+x}\right) dx + dy = 0$$

Then solve the exact equation that results.

Problem 7: Find A y -Only Integrating Factor. Find an integrating factor depending only on y for:

$$(1 + x^2) dx + y \left(x + \frac{x^3}{3}\right) dy = 0$$

Then give the potential function.

Problem 8: Diagnose A Failed Test. For:

$$(x + y^2) dx + (x^2 + y) dy = 0,$$

calculate both one-variable integrating-factor test quotients. Explain why neither standard test applies directly.

Problem 9: Check A Proposed Potential. Determine whether:

$$\Phi(x, y) = x^2 e^y + \ln(1 + y^2)$$

is a potential for:

$$2xe^y dx + \left(x^2 e^y + \frac{2y}{1 + y^2} \right) dy = 0.$$

Problem 10: Check A Proposed Integrating Factor. Determine whether $\mu(y) = e^{2y}$ is an integrating factor for:

$$(3x + 1) dx + \left(\frac{3}{2}x^2 + x + e^{-2y} \right) dy = 0.$$

Practice Answers

Answer 1. Identify:

$$M = y^2 + x, \quad N = 2xy + e^y$$

Differentiate:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(y^2 + x) \\ &= 2y \end{aligned}$$

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(2xy + e^y) \\ &= 2y \end{aligned}$$

The cross-partial agree, so the equation is exact on the plane.

Answer 2. Identify the coefficients:

$$M = 2xy + \cos x, \quad N = x^2 + 3y^2$$

Test exactness:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y} (2xy + \cos x) \\ &= 2x \end{aligned}$$

$$\begin{aligned} N_x &= \frac{\partial}{\partial x} (x^2 + 3y^2) \\ &= 2x \end{aligned}$$

Now integrate M with respect to x :

$$\begin{aligned} \Phi &= \int (2xy + \cos x) dx + h(y) \\ &= x^2y + \sin x + h(y). \end{aligned}$$

Differentiate with respect to y :

$$\Phi_y = x^2 + h'(y)$$

Compare with $N = x^2 + 3y^2$:

$$x^2 + h'(y) = x^2 + 3y^2$$

Subtract x^2 from both sides:

$$h'(y) = 3y^2$$

Integrate with respect to y :

$$\begin{aligned} h(y) &= \int 3y^2 dy \\ &= y^3. \end{aligned}$$

Substitute $h(y) = y^3$ into the reconstructed potential:

$$\Phi(x, y) = x^2y + \sin x + y^3$$

The exact equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so the solution is:

$$\boxed{x^2y + \sin x + y^3 = C}$$

Answer 3. Here:

$$M = e^x + y, \quad N = x + 2y$$

Differentiate each coefficient across the variables:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(e^x + y) \\ &= 1 \end{aligned}$$

and:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(x + 2y) \\ &= 1 \end{aligned}$$

The cross-partial agree, so the form is exact.

Integrate N with respect to y :

$$\begin{aligned} \Phi &= \int (x + 2y) dy + k(x) \\ &= xy + y^2 + k(x). \end{aligned}$$

Differentiate with respect to x :

$$\Phi_x = y + k'(x)$$

Compare with $M = e^x + y$:

$$y + k'(x) = e^x + y$$

Subtract y from both sides:

$$k'(x) = e^x$$

Integrate with respect to x :

$$\begin{aligned} k(x) &= \int e^x dx \\ &= e^x. \end{aligned}$$

Substitute $k(x) = e^x$ into the reconstructed potential:

$$\Phi(x, y) = xy + y^2 + e^x$$

Since the exact equation is $d\Phi = 0$, integrate to obtain $\Phi = C$:

$$\boxed{xy + y^2 + e^x = C}$$

Answer 4. Multiply by $x + 2y$:

$$(x + 2y)y' = -(y + 2x)$$

Move all terms to the left:

$$(y + 2x) + (x + 2y)y' = 0$$

Replace y' by dy/dx :

$$(y + 2x) + (x + 2y)\frac{dy}{dx} = 0$$

Multiply every term by dx :

$$(y + 2x) dx + (x + 2y) dy = 0$$

The form is exact because:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(y + 2x) \\ &= 1 \end{aligned}$$

and:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x}(x + 2y) \\ &= 1. \end{aligned}$$

Integrate M with respect to x :

$$\begin{aligned} \Phi &= \int (y + 2x) dx + h(y) \\ &= xy + x^2 + h(y), \end{aligned}$$

where y is held constant.

Then:

$$\Phi_y = x + h'(y)$$

Compare with $N = x + 2y$:

$$x + h'(y) = x + 2y$$

Subtract x from both sides:

$$h'(y) = 2y$$

Integrate with respect to y :

$$\begin{aligned}h(y) &= \int 2y \, dy \\ &= y^2.\end{aligned}$$

Substitute $h(y) = y^2$ into the reconstructed potential:

$$\Phi(x, y) = xy + x^2 + y^2$$

The exact equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so the solution is:

$$\boxed{x^2 + xy + y^2 = C}$$

Answer 5. Use the potential:

$$\Phi(x, y) = xy + y^2 + e^x$$

Check it locally:

$$\Phi_x = y + e^x = M$$

and:

$$\Phi_y = x + 2y = N$$

Both coefficients are recovered, so the differential equation is $d\Phi = 0$. Integrating gives $\Phi = C$, or:

$$xy + y^2 + e^x = C$$

Apply $(0, 1)$:

$$(0)(1) + (1)^2 + e^0 = C$$

Therefore:

$$0 + 1 + 1 = C$$

so:

$$C = 2$$

The initial-value solution is:

$$\boxed{xy + y^2 + e^x = 2}$$

Answer 6. Identify:

$$M = 2 + \frac{3y}{1+x}, \quad N = 1$$

Then:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y} \left(2 + \frac{3y}{1+x} \right) \\ &= \frac{3}{1+x} \end{aligned}$$

$$\begin{aligned} N_x &= \frac{\partial}{\partial x} (1) \\ &= 0 \end{aligned}$$

The x -test gives:

$$\begin{aligned} \frac{M_y - N_x}{N} &= \frac{\frac{3}{1+x} - 0}{1} \\ &= \frac{3}{1+x} \end{aligned}$$

Therefore:

$$\begin{aligned}\mu(x) &= \exp\left(\int \frac{3}{1+x} dx\right) \\ &= \exp(3 \ln |1+x|) \\ &= |1+x|^3.\end{aligned}$$

On $x > -1$, this is $(1+x)^3$. On $x < -1$, it is $-(1+x)^3$. These differ only by the nonzero constant factor -1 , so on either interval we may use the simpler representative:

$$\boxed{\mu(x) = (1+x)^3}$$

Multiply the equation:

$$\left[2(1+x)^3 + 3y(1+x)^2\right] dx + (1+x)^3 dy = 0$$

Call the transformed coefficients $\widetilde{M}$ and $\widetilde{N}$:

$$\widetilde{M} = 2(1+x)^3 + 3y(1+x)^2$$

$$\widetilde{N} = (1+x)^3$$

Retest exactness:

$$\widetilde{M}_y = 3(1+x)^2$$

and:

$$\widetilde{N}_x = 3(1+x)^2$$

The transformed equation is exact.

Integrate the dy coefficient with respect to y :

$$\begin{aligned}\Phi &= \int (1+x)^3 dy + k(x) \\ &= (1+x)^3 y + k(x),\end{aligned}$$

where x is held constant during the integration.

Differentiate with respect to x :

$$\Phi_x = 3(1+x)^2 y + k'(x)$$

Set this equal to $\widetilde{M}$:

$$3(1+x)^2 y + k'(x) = 2(1+x)^3 + 3y(1+x)^2$$

Subtract the common $3y(1+x)^2$ term:

$$k'(x) = 2(1+x)^3$$

Integrate using $u = 1+x$, so $du = dx$:

$$\begin{aligned}k(x) &= \int 2(1+x)^3 dx \\ &= \int 2u^3 du \\ &= \frac{u^4}{2} \\ &= \frac{(1+x)^4}{2}.\end{aligned}$$

Substitute this $k(x)$ into the reconstructed potential:

$$\Phi(x, y) = (1+x)^3 y + \frac{(1+x)^4}{2}$$

The transformed exact equation is $d\Phi = 0$. Integrating gives $\Phi = C$, so the solution is:

$$(1+x)^3y + \frac{(1+x)^4}{2} = C$$

Answer 7. Identify:

$$M = 1 + x^2$$

$$N = y \left(x + \frac{x^3}{3} \right)$$

Compute:

$$\begin{aligned} M_y &= \frac{\partial}{\partial y}(1 + x^2) \\ &= 0 \end{aligned}$$

and:

$$\begin{aligned} N_x &= \frac{\partial}{\partial x} \left[y \left(x + \frac{x^3}{3} \right) \right] \\ &= y(1 + x^2). \end{aligned}$$

The y -test gives:

$$\frac{N_x - M_y}{M} = \frac{y(1 + x^2)}{1 + x^2} = y$$

Therefore:

$$\begin{aligned} \mu(y) &= \exp \left(\int y \, dy \right) \\ &= \boxed{e^{y^2/2}}. \end{aligned}$$

Multiply both original coefficients by $e^{y^2/2}$:

$$e^{y^2/2}(1+x^2)dx + e^{y^2/2}y\left(x + \frac{x^3}{3}\right)dy = 0$$

The transformed coefficients are:

$$\widetilde{M} = e^{y^2/2}(1+x^2)$$

$$\widetilde{N} = e^{y^2/2}y\left(x + \frac{x^3}{3}\right)$$

Retest exactness. Differentiating $\widetilde{M}$ with respect to y gives:

$$\widetilde{M}_y = ye^{y^2/2}(1+x^2)$$

Differentiating $\widetilde{N}$ with respect to x gives:

$$\widetilde{N}_x = ye^{y^2/2}(1+x^2)$$

The transformed equation is exact.

Now integrate $\widetilde{M}$ with respect to x . The factor $e^{y^2/2}$ is constant during this integration:

$$\begin{aligned}\Phi &= \int e^{y^2/2}(1+x^2)dx + h(y) \\ &= e^{y^2/2} \int (1+x^2)dx + h(y) \\ &= e^{y^2/2} \left(x + \frac{x^3}{3}\right) + h(y).\end{aligned}$$

Differentiate with respect to y :

$$\Phi_y = ye^{y^2/2} \left(x + \frac{x^3}{3}\right) + h'(y)$$

Compare this with $\tilde{N}$:

$$ye^{y^2/2} \left(x + \frac{x^3}{3} \right) + h'(y) = ye^{y^2/2} \left(x + \frac{x^3}{3} \right)$$

Subtract the common term:

$$h'(y) = 0$$

Thus $h(y)$ is constant, and that constant may be omitted from the potential:

$$\Phi(x, y) = e^{y^2/2} \left(x + \frac{x^3}{3} \right)$$

Answer 8. Here:

$$M = x + y^2, \quad N = x^2 + y$$

Then:

$$M_y = 2y, \quad N_x = 2x$$

The x -test quotient is:

$$\frac{M_y - N_x}{N} = \frac{2y - 2x}{x^2 + y},$$

which depends on both variables.

The y -test quotient is:

$$\frac{N_x - M_y}{M} = \frac{2x - 2y}{x + y^2},$$

which also depends on both variables.

Therefore neither one-variable integrating-factor test applies directly.

Answer 9. Differentiate the proposed potential:

$$\Phi_x = 2xe^y$$

For the y derivative:

$$\begin{aligned}\Phi_y &= x^2e^y + \frac{1}{1+y^2}(2y) \\ &= x^2e^y + \frac{2y}{1+y^2}.\end{aligned}$$

These derivatives match the two displayed coefficients, so Φ is a potential.

Answer 10. Multiply the equation by e^{2y} :

$$e^{2y}(3x+1)dx + \left[e^{2y}\left(\frac{3}{2}x^2 + x\right) + 1 \right]dy = 0$$

Let the transformed coefficients be $\tilde{M}$ and $\tilde{N}$.

Differentiate:

$$\tilde{M}_y = 2e^{2y}(3x+1)$$

and:

$$\tilde{N}_x = e^{2y}(3x+1).$$

These are not equal. Therefore the proposed factor e^{2y} is **not** an integrating factor.

The mismatch is deliberate: it demonstrates that a proposed factor must be verified rather than accepted from appearance alone.

To find the correct factor, return to the original coefficients. Their cross-partials are:

$$M_y = 0$$

Differentiate N with respect to x :

$$N_x = 3x + 1$$

Therefore, where $3x + 1 \neq 0$:

$$\begin{aligned}\frac{N_x - M_y}{M} &= \frac{(3x + 1) - 0}{3x + 1} \\ &= 1.\end{aligned}$$

Integrating this result gives:

$$\begin{aligned}\mu(y) &= \exp\left(\int 1 \, dy\right) \\ &= e^y.\end{aligned}$$

Thus a valid integrating factor is:

$$\boxed{\mu(y) = e^y}$$

To verify it directly, multiply the original coefficients by e^y :

$$\widehat{M} = e^y(3x + 1)$$

$$\widehat{N} = e^y\left(\frac{3}{2}x^2 + x\right) + e^{-y}$$

Then:

$$\widehat{M}_y = e^y(3x + 1)$$

and:

$$\widehat{N}_x = e^y(3x + 1)$$

The cross-partials agree everywhere. This direct check also shows that the factor remains valid on the line $3x + 1 = 0$, where the discovery quotient was undefined.

Final Takeaway

Exact equations are solved by uncovering a potential function:

$$M dx + N dy = d\Phi$$

The test:

$$M_y = N_x$$

checks whether that potential can exist on the working domain.

The reconstruction then uses one partial integration and one coefficient comparison. If the original form is not exact, a one-variable integrating factor may repair it.

The main lesson is:

show each partial derivative, integration choice, comparison, and substitution clearly enough that every line can be checked against the original equation.